

1 Overview

In this assignment, you will plan, design, implement, and deploy a simple web service. You will first write a Product Requirements Document (PRD) that clearly explains what service you want to build, who will use it, what problem it solves, and what features it should provide. Based on your PRD, you will then create a working website with the help of AI programming tools and deploy it using Vercel.

The goal of this assignment is not only to make a webpage, but also to experience the full product development process: **idea definition, service planning, requirement writing, AI-assisted implementation, code organization, and online deployment.**

2 Assignment Objectives

Through this assignment, students will learn how to:

- Define a simple but meaningful web service idea.
- Write a clear and actionable Product Requirements Document (PRD).
- Translate product requirements into an actual website implementation.
- Use AI tools effectively for web development while understanding and verifying the generated code.
- Deploy a web service publicly using Vercel.
- Explain the relationship between service planning, implementation, and final user experience.

3 Assignment Theme

You may choose your own web service topic. Your service does not need to be technically complex, but it should have a clear purpose and target users.

Possible examples include:

- A personal portfolio website.
- A simple study planner for students.
- A course information or assignment tracking website.
- A restaurant, cafe, or travel recommendation page.
- A simple habit tracker or productivity tool.
- A small AI-themed educational website.
- A landing page for a fictional startup or service.

Your project should be designed as a **service**, not just a collection of static pages. This means that your website should clearly communicate what value it provides to users and how users are expected to interact with it.

4 Assignment Requirements

4.1 Step 1: PRD Document (40%)

You must write a PRD before implementing your website. The PRD should be submitted as a well-organized PDF document.

Your PRD must include the following components:

- **Product Overview:** Clearly describe the product or service you are building. Explain the main idea, target users, and the problem or need your service addresses.
- **Target Users:** Define who will use your service. Describe their needs, pain points, or expected usage scenarios.
- **Project Goals:** State the main goals of the project. Goals should be specific and realistic.
- **Core User Scenario:** Describe at least one realistic scenario showing how a user would use your website.
- **Feature List:** List the main features of your website. Each feature should include a short description and priority level such as Must-have, Should-have, or Nice-to-have.
- **Page Structure:** Describe the pages or major sections of your website. For example: Home, About, Features, Contact, Dashboard, or Planner.
- **Technical Requirements:** Specify the technologies you plan to use, such as HTML, CSS, JavaScript, React, Next.js, or other frameworks. You may use a simple static website or a framework-based website.
- **Design Requirements:** Describe the intended visual style, layout, color direction, navigation structure, and user experience principles.
- **Milestones:** Provide a simple development plan with key milestones, such as PRD completion, first prototype, final implementation, and Vercel deployment.

The PRD does not need to be long, but it must be clear, structured, and actionable. A good PRD should make it easy to understand what you are trying to build and how the final website should behave.

4.2 Step 2: Website Implementation (45%)

You must build a working website based on your PRD. You may use HTML, CSS, and JavaScript, or a modern web framework such as React or Next.js.

Your website must satisfy the following minimum requirements:

- Include at least **three pages** or three clearly separated sections.
- Provide a clear navigation structure.
- Use appropriate layout and styling with HTML and CSS.
- Include at least one meaningful interactive feature using JavaScript.
- Be consistent with the product idea, target users, and feature list described in your PRD.
- Be deployed online using **Vercel**.

Examples of acceptable interactive features include:

- A form that collects user input and displays a result.
- A to-do list or planner feature.
- A simple recommendation or filtering function.
- A button-based interaction that changes page content.
- A quiz, calculator, checklist, or progress tracker.
- A modal window, tab menu, slider, or dynamic card list.

The interaction should support the purpose of your service. For example, a dark mode toggle alone is not sufficient unless it is clearly connected to the service design.

4.3 Step 3: AI-Assisted Development and Deployment Report (15%)

You are encouraged to use AI tools such as ChatGPT, Claude, Gemini, GitHub Copilot, or other AI coding assistants. However, you are responsible for understanding, testing, and explaining your final code.

You must include a short AI development report in your submission. This report may be included in the PRD PDF or submitted as a separate PDF.

The report must include:

- Which AI tools you used.
- What tasks you asked AI to help with.
- At least three representative prompts or instructions you used.
- What parts of the AI-generated result you modified or improved.
- Any bugs, errors, or limitations you found and how you fixed them.
- The final Vercel deployment URL.

Simply copying AI-generated code without understanding it is not acceptable. You may be asked to explain your code and design decisions during evaluation.

5 Submission Guidelines

You must submit the following items:

1. **PRD Document:** A clear and concise PDF file.
2. **AI Development Report:** Included in the PRD PDF or submitted as a separate PDF.
3. **GitHub Repository Link:** The repository must include all source code files.
4. **Vercel Deployment Link:** The final website must be accessible online.
5. **README File:** Your GitHub repository must include a README file explaining the project overview, main features, how to run the project locally, and the deployment URL.

Your project folder should be organized clearly. For example:

- `index.html`, `style.css`, `script.js` for a simple static website.
- Or a framework-based structure such as `src/`, `app/`, `components/`, and `public/`.

6 Important Notes

- Your final website must match the direction described in your PRD.
- You do not need to implement a backend server or database.
- You may use templates or AI-generated code, but you must customize them for your own service idea.
- Broken links, missing files, or non-working deployment may result in grade deductions.
- The website does not need to be complex, but it should be complete, coherent, and usable.
- You should be able to explain your service concept, PRD decisions, and source code.

7 Expected Outcome

By the end of this assignment, you should have:

- A complete PRD document describing your web service.
- A working website implemented according to your PRD.
- A GitHub repository containing your project source code.
- A publicly accessible Vercel deployment link.
- Practical experience using AI as a programming assistant in a real development workflow.